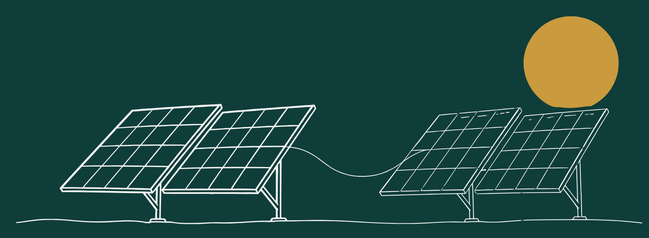
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A Trustworthy Agentic Digital Twin for Adaptive Solar Energy Management

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Abstract

Forecasting models trained at one solar plant often lose accuracy at a new site. Our accepted NEECON 2026 paper measured both outcomes: successful transfer within a region and negative transfer across regions. **SolarTwin AI** is a software digital twin that checks transfer risk, compares forecast models, plans battery and flexible-load actions within hard limits, and records why each action was taken.

Problem Statement & Objectives

New plants have little generation history. A model borrowed from another plant can drift without warning, and an energy schedule built on that forecast inherits the error.

1. **Detect** when a transferred forecast should not be trusted.
2. **Decide** a safe battery and flexible-load schedule within operating limits.
3. **Explain** each decision with the evidence that produced it.

Related Literature

Solar forecasting usually depends on site-specific history. Domain adaptation (DANN, CDAN) aligns source and target features so a model can transfer without target labels; our ISGT Middle East 2025 study reached $R^2 = 0.75$ in blind cross-plant transfer [2]. Digital twins let operators test plant and storage decisions in simulation before acting on them.

Design & Validation

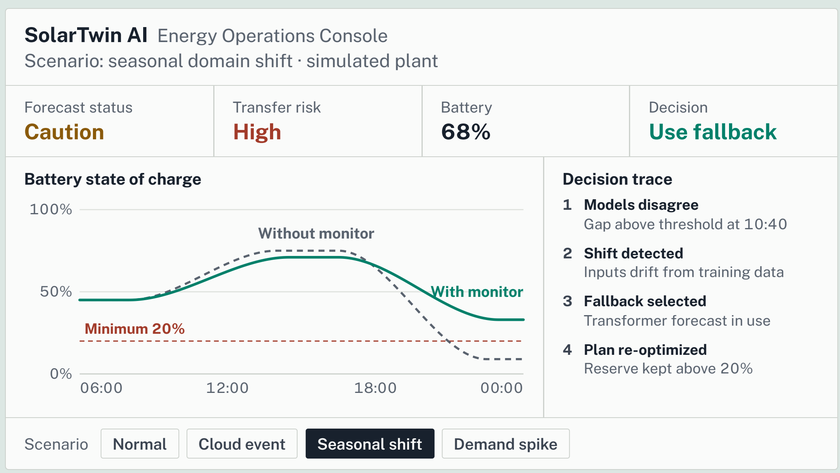


Fig. 2. Energy Operations Console. Judges pick a scenario and watch the forecast, risk signal and battery plan respond. Simulated plant and data.

Methodology

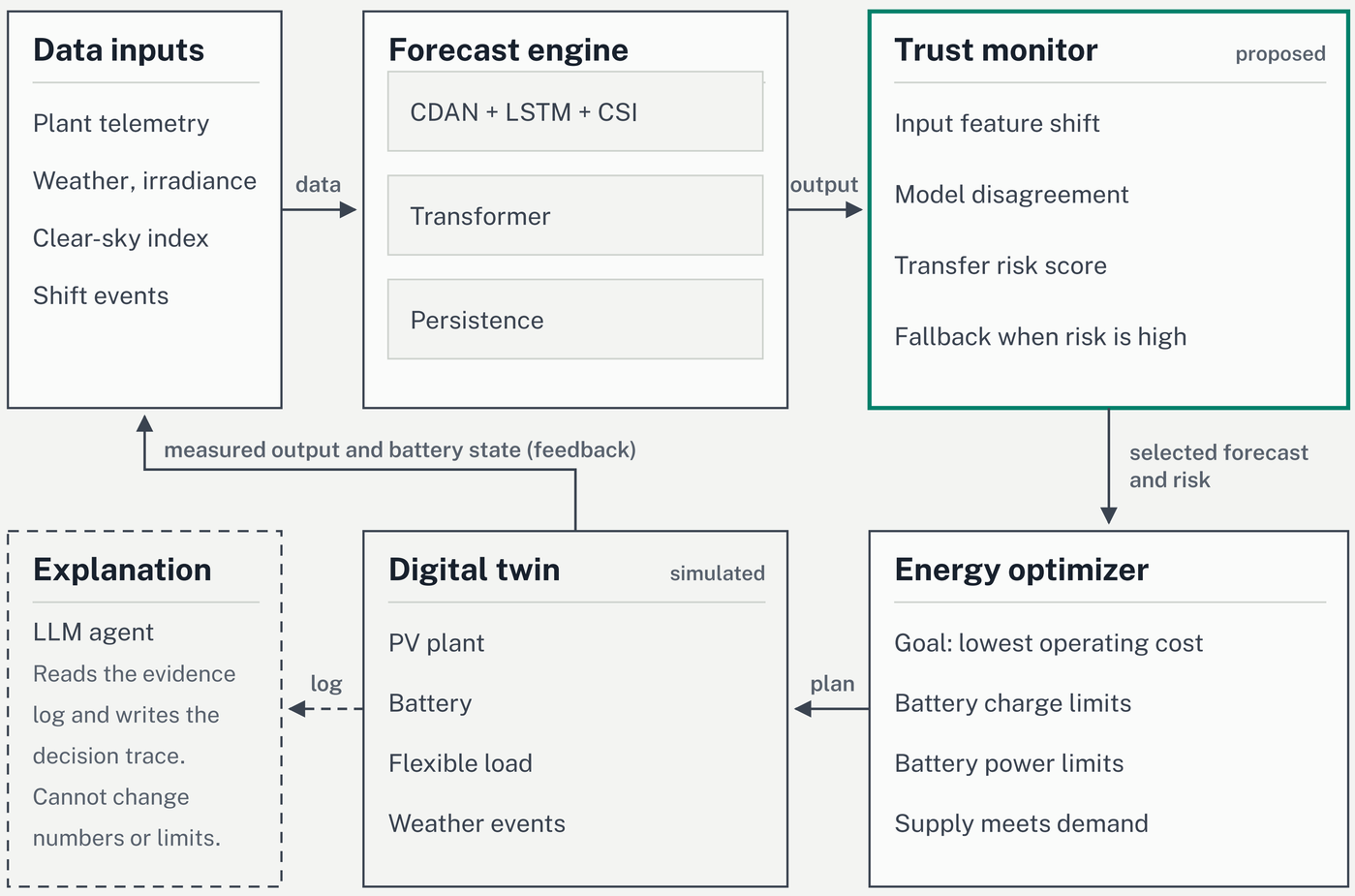


Fig. 1. System architecture. Solid arrows are numerical signals computed in code; the dashed arrow is the evidence log the LLM agent reads to explain decisions.

Results & Analysis

Transfer case [1]	Model	Result
Plant 1 to Plant 2 Same region, capacity mismatch	CDAN + LSTM + CSI	**$R^2 = 0.8325$**
India to Australia Cross-regional	CDAN + LSTM + CSI	**Negative transfer**
India to Australia Cross-regional	Transformer No adaptation	**Positive skill**

Adaptation helps when sites are similar and can hurt when they differ, so transfer risk must be measured before a forecast drives a decision. Fig. 3 shows how the monitor responds in a simulated shift.

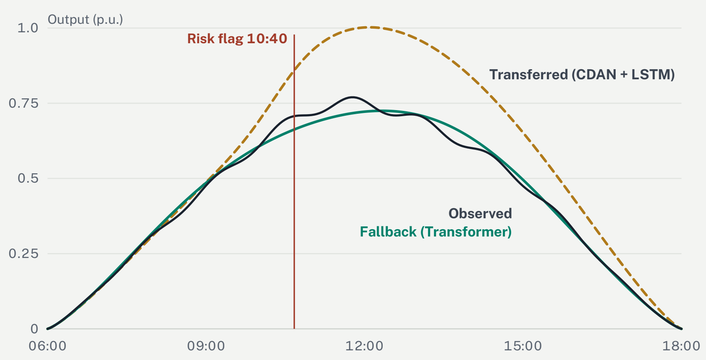


Fig. 3. Forecast candidates against observed output in a simulated seasonal shift. After the risk flag, the fallback forecast is used.

Evaluation Protocol



Forecast
MAE
RMSE
R^2

Risk monitor
Precision
Recall
False alarms

Energy plan
Operating cost
Renewable use
Unmet demand

Conclusion

Takeaway
Forecast trust can be measured and acted on before it affects operations.

Contribution to theory
Links capacity-aware transfer learning with risk monitoring and constrained control.

Contribution to practice
A low-cost, offline tool to test new sites, weather events and demand changes.

Future work
Report detector precision, recall and false alarms; compare cost and unmet demand with baselines.

References & Contact

[1] Belay et al., “Learning When to Transfer: Robust Solar Power Forecasting Under Capacity Mismatch,” accepted at NEECON 2026.
[2] D. E. Belay et al., “Domain Adaptation for Cross-Plant Solar Power Forecasting Using Time Series Deep Learning Models,” IEEE PES ISGT Middle East 2025.